From Static to Contextual: Distributional Semantics and What's Inside BERT

Combined report: two independent implementations compared
NLP CS60075 Assignment II – Project 5, dataset: UD English EWT

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Abstract

The two of us implemented the project independently and compare the two implementations here: **A** (Y. D. Babu) and **B** (Trishita Saha). The project builds PPMI+SVD vectors from UD English EWT (Task 1), reproduces the Hewitt and Manning (2019) structural probe on `bert-base-uncased` (Task 2), and pools the most syntactic layer into static vectors (Task 3). **Both implementations find the same best layer (ℓ^*=8) and the same qualitative picture.** BERT-static vectors are far better than PPMI-SVD on WordSim-353 and analogies, but no better (in fact worse) on the tree-recovery check, so the semantic and syntactic evaluations disagree. Under identical settings the two Task 3 pipelines give *the same numbers to three decimals*. The one large quantitative difference is the probe's UUAS (A 79.5%, B 68.9%, paper 79.8%). We trace it to the training schedule: B uses a constant learning rate, A uses the paper's decay. A control run in A with B's schedule shows the same drop (62.0% when the last epoch is kept), and the schedule also changes which probe rank looks best.

1 Introduction

Count-based distributional models give one fixed vector per word (Levy and Goldberg, 2014); BERT (Devlin et al., 2019) gives every token a context-dependent vector. On UD English EWT (Silveira et al., 2014; de Marneffe et al., 2021) the project asks:

- **T1:** how good are PPMI+SVD vectors from 200k words?
- **T2:** which BERT layer encodes dependency trees best, according to a structural probe?
- **T3:** if that layer is averaged into static vectors, are they better than T1 on meaning and on tree structure – and do the two answers agree?
- **Ours, because we implemented everything twice:** which conclusions survive a change of implementation?

Summary of the comparison

- **Robust (both implementations agree):**
  - ℓ^*=8 with a mid-layer peak;
  - T3 $\gg$ T1 on WordSim-353 and analogies;
  - T3 < T1 on the MST check, close to a random tree;
  - the best layers for static semantics are 1–2, not ℓ^*;
  - mean-centring / all-but-the-top helps T3.
- **Implementation-sensitive:**
  - the absolute probe UUAS and the best probe rank (training schedule);
  - T1's WordSim ρ (vocabulary cut-off and pair subset);
  - absolute MST levels (sentence sample and punctuation handling).

2 Data and the two implementations

Shared by both implementations

- EWT splits: 12,544 / 2,001 / 2,077 sentences; 204,578 train tokens; 16,654 lowercased types.
- Benchmarks: WordSim-353 (Finkelstein et al., 2002) and Google analogies (Mikolov et al., 2013), gensim copies.
- Model: frozen `bert-base-uncased`; a word's vector = mean of its word pieces.
- The paper's probes and L1 losses; ℓ^* selected on dev.

Table 1 lists every choice in which the two implementations differ.

	A (Y. D. Babu)	**B** (Trishita Saha)
EWT loading	HF `universal_dependencies`, pinned tag `ud2.18-loader-v2.3.0`	official `.conllu` files, GitHub `master` branch (HF loader failed for B)
T1 vocabulary	all 16,654 types (min. count 1)	8,865 types with count ≥2; rarer words dropped
SVD	sklearn `TruncatedSVD`, randomized, seed 42	scipy `svds`, fixed start vector
WS-353 "all 353"	OOV pair → median model similarity	OOV pair → similarity 0
Analogy coverage	5,727 covered questions	3,406 covered questions
20-sentence sample	seed 42; 5–40 non-punct. words, *all in vocabulary*; punctuation excluded (234 edges)	seed 42; ≥5 tokens, any words; *punctuation included*; OOV words at distance 1 (282 edges)
Random-tree baseline	uniformly random spanning tree (expected UUAS)	MST of random distances (100 draws)
Robustness set	all 660 eligible test sentences	all 1,926 multi-token test sentences
Probed layers	0 (embeddings) + 1–12; 78 probes	1–12; 72 probes
Probe schedule	Adam 1e-3, 30 ep., **lr ×0.1 when dev loss stalls** (paper), best-dev checkpoint	Adam 1e-3, 30 ep., **constant lr**, best-dev checkpoint
UUAS punctuation	excluded (paper convention)	reported with and without
Depth Spearman	per-length average, macro over lengths 5–50 (paper)	per-sentence mean over lengths 5–50
T3 vocabulary	all 16,654 types	main: T1's 8,865 types; variant: all types
Anisotropy fix	all-but-the-top, $D = 7$ (Mu and Viswanath, 2018); also on T1	mean-centring; centring + PCA to 100 dims
Uncertainty	bootstrap CIs, paired bootstrap / McNemar tests	micro and macro averages; full test split
Seeds	42 everywhere; probe $42 + 1000\ell + k$ (+500 depth)	sample 42; SVD start 0; probe $1000k$ (+7 depth) + layer

Table 1: Where implementations A and B differ (everything else is shared). B's entries were checked against B's source code (`Codes/*.py`).

Cross-check: same settings give the same numbers (Table 2)

- B's raw outputs are not available to us, so A's outputs were re-scored under B's settings.
- Every Task 3 number of B that can be reproduced this way is reproduced **exactly**: the two pooling and evaluation pipelines are equivalent.
- The T1 numbers differ only because B's vectors were trained on the count-≥2 vocabulary.

quantity	A (recomputed in B's setting)	B (reported)	
train / dev / test sentences	12,544 / 2,001 / 2,077	12,544 / 2,001 / 2,077	identical
train tokens; lowercased types	204,578; 16,654	204,578; 16,654	identical
types with count ≥2 (B's T1 vocabulary)	8,865	8,865	identical
analogy questions, all words count ≥2	3,406	3,406	identical
T3 layer 8, WS-353 ρ on those 203 pairs	0.648	0.648	identical
T3 layer 8 (all types), WS ρ, 248 covered pairs	0.577	0.577	identical
T3 layer 8 (all types), WS ρ, all 353, missing = 0	0.288	0.288	identical
T3 layer 8 (all types), analogy, 5,727 covered	67.7	67.7	identical
semantic / syntactic	74.5 / 66.6	74.5 / 66.6	identical
T3 layer 8 (all types), analogy, all 19,544	19.8	19.8	identical
T1, WS ρ on the 203 pairs	0.171^a	0.189^b	vocabulary cut-off
T1, WS ρ, all 353, missing = 0	0.060^a	0.024^b	vocabulary cut-off

Table 2: A's outputs re-scored in B's setting vs. B's reported numbers. ^{a}A's vectors (min. count 1); ^{b}B's vectors (count ≥2), so the T1 rows compare two different vector sets on the same pairs.

3 Task 1 – PPMI + SVD

Method (both)

- Symmetric ±5-word co-occurrence matrix over the training sentences; PPMI (Levy and Goldberg, 2014); top 100 SVD dimensions, vectors $U\Sigma$.
- Cosine similarity; analogies by the vector offset with the three query words excluded.

Task 1	A	B
vocabulary	all 16,654 train types	8,865 types (count ≥2)
WS-353 ρ, covered pairs	0.104 (248)	0.189 (203)
WS-353 ρ, all 353	0.048 (median) / 0.060 (zero)	0.024 (zero)
analogy acc., covered	0.51% (5,727)	0.85% (3,406)
semantic / syntactic	2.1 / 0.26%	2.0 / 0.70%
analogy acc., all 19,544	0.15%	0.15%
MST UUAS, 20 sentences	20.5 (234 edges)	16.7 (282 edges)
random tree / linear chain	15.2 / 43.2	13.1 / 35.5
MST UUAS, robustness set	24.8 (660 sent.)	17.8 (1,926 sent.)
random tree / linear chain	17.1 / 43.0	14.3 / 40.5

Table 3: Task 1 in both implementations (numbers in brackets: covered items / sentences or edges).

Agreement

- Both find weak, data-limited vectors: WS-353 ρ near 0 over all 353 pairs and analogies below 1%.
- In both, the MST beats a random tree but stays well below the linear chain.

Differences explained

1. **B's higher covered** ρ (0.189 vs. 0.104) has two causes:
   - *different pairs* – B's 203 covered pairs exclude words seen once; A's vectors score 0.171 on those same pairs, so most of the gap is the subset (consistent with A's frequency analysis: ρ 0.317 on pairs of words seen ≥20 times);
   - *different vectors* – dropping singletons removes noisy rows and contexts before the SVD (0.171 → 0.189).
2. **"All 353" numbers** differ by the imputation rule: with B's "missing = 0" rule A gets 0.060 (B 0.024); with the median rule A gets 0.048.
3. **MST levels** differ because B's sample keeps OOV words (at distance 1) and punctuation, which lowers every system and both baselines (B: random 13.1, chain 35.5; A: 15.2, 43.2). The ordering random < T1 < chain is the same in both.

Extra analyses (A)

- **Levy et al. hyper-parameters** (Levy et al., 2015) (Table 4): no setting rescues the analogies; the best variant ($p = 0$) only reaches $\rho = 0.134$ – the limit is the 200k-word corpus.
- **Per relation:** PPMI-SVD recovers content-word relations that co-occur (amod 0.40, conj 0.34) and almost no function-word attachments (case 0.12, det 0.14).

	WordSim-353 ρ			Analogy acc. (%)		MST UUAS (%)	
PPMI-SVD variant	covered (248) [95% CI]	all 353	freq. ≥20	covered	all	20 sent.	pool (660)
spec: α=1, p=1 (UΣ)	0.104 [−0.03, 0.22]	0.048	0.317	0.51	0.15	20.5	24.8
α=0.75, p=1	0.060 [−0.07, 0.20]	0.043	0.199	0.65	0.19	15.8	20.8
α=1, p=0.5	0.129 [−0.00, 0.24]	0.091	0.289	0.37	0.11	20.9	24.6
α=1, p=0 (U)	0.134 [0.00, 0.25]	0.089	0.320	0.21	0.06	22.6	24.5
α=0.75, p=0.5	0.097 [−0.04, 0.22]	0.066	0.174	0.42	0.12	16.2	20.8
spec + all-but-the-top (D=1)	0.153	0.103	0.294	0.42	–	20.5	24.5
linear chain (attach to next word)	–	–	–	–	–	43.2	43.0
random tree (expected)	–	–	–	–	–	15.2	17.1

Table 4: A's Task 1 with the Levy et al. variants (α = context-distribution smoothing, p = eigenvalue weight) and baselines.

4 Task 2 – Structural probe

Method (both)

- All layers of frozen `bert-base-uncased` extracted once (word = mean of its word pieces; fp16 cache).
- Distance probe $\|B(h_i - h_j)\|^2$ and depth probe $\|Bh_i\|^2$; $B \in \mathbb{R}^{k \times 768}$, $k \in \{32, 64, 128\}$, $B \sim \mathcal{U}(-0.05, 0.05)$; L1 loss normalised by n^2 / n.
- Adam lr 10^{-3}, batches of 20 sentences, 30 epochs, best-dev checkpoint; MST decoding; ℓ^* on dev UUAS.
- **The decisive difference is the learning-rate schedule** (Table 1).

Agreement

- Both pick ℓ^*=8.
- Both see UUAS and depth Spearman rise to the middle layers and fall towards layer 12 (Fig. 1).
- In both, the dev choice equals the test argmax or is statistically tied with it (A: layers 6–8 are tied by a paired bootstrap).

The UUAS gap is the schedule (A 79.5%, B 68.9%, paper 79.8%)

1. **Control run:** A retrained with B's schedule (constant lr, last epoch, $k = 128$) falls to 62.0%; B, with the same schedule but best-dev selection, sits in between (68.9%, Fig. 1).
2. **Same symptom:** B observes that the best dev epoch of $k = 128$ distance probes is mostly epoch 1; in A's control the dev minimum is also at epoch 1, after which dev loss oscillates (0.54–0.59) for 29 epochs.
3. **Not over-fitting but non-convergence:** in A's control the *training* loss also stays high (0.62 vs. 0.42 with decay). With an L1 loss the gradient does not shrink near the optimum, so Adam's steps stay ≈lr and circle the minimum; the ×0.1 decay lets the probe settle.

The schedule also reverses the rank conclusion

- B (constant lr) finds $k = 32$ best and reads it as over-fitting of larger probes.
- A (decay) finds $k = 128 \geq 64 > 32$ (at ℓ^*: 77.4 / 79.2 / 79.5%), as the paper does.
- The smaller probe is simply less hurt by non-convergence, so "low rank generalises better" is a property of the schedule, not of the data.
- Both agree that the tree lives in a low-dimensional subspace: even $k = 32$ is within about 2 pt of $k = 128$ in A.

Remaining gap to the paper (A, −0.3 pt) and other observations

- **Plausible causes (named by both):** 12.5k vs. ≈40k training sentences; web text vs. edited news; UD content-head trees vs. Stanford Dependencies; uncased vs. cased BERT; rank ≤128 and 30 epochs vs. full rank and up to 40 epochs.
- **What context adds (A):** the layer-0 probe (58.7%) matches the paper's non-contextual PROJ0 (59.8%), so context adds ≈21 pt, mostly for function words (case 0.35→0.73, cc 0.40→0.82).
- **Easy and hard relations (A, Fig. 2):** short head–dependent arcs are easy (obj 0.93, det 0.91, aux 0.89); coordination (conj 0.56) and PP attachment (obl 0.69) are hard – UD attaches every conjunct to the first, creating long arcs.
- **Punctuation (B):** including it costs 2.5 pt (66.4 vs. 68.9), because UD attaches punctuation to content heads.
- **Root accuracy:** A 87.5%, B 72.4% (B: 48–57% for $k = 128$), again consistent with the schedule effect.

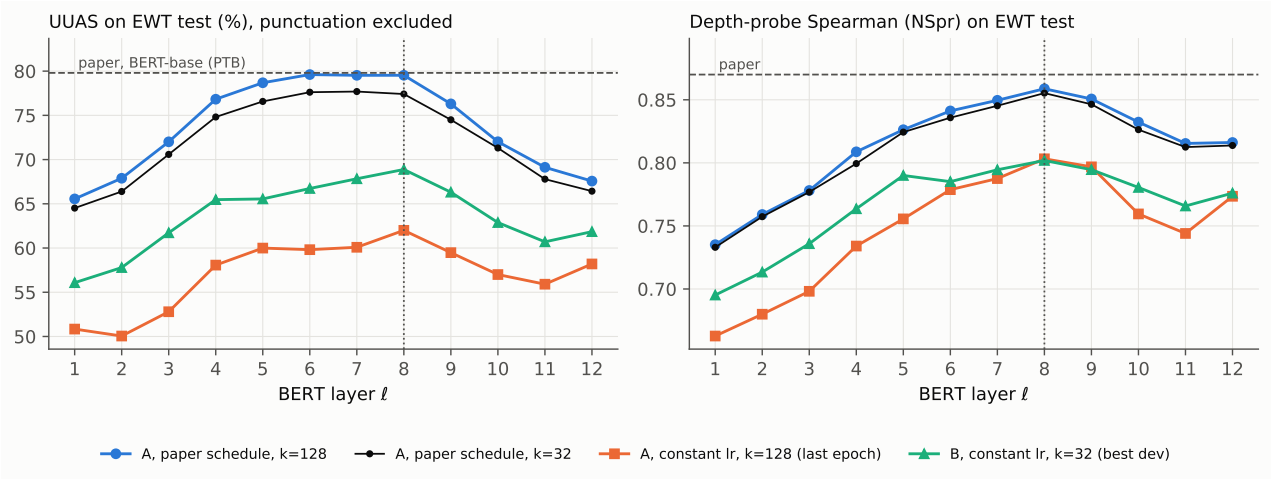


Figure 1: Probe quality vs. layer in both implementations (EWT test). A's curves come from its outputs; B's from its Table 2. A's constant-lr control (orange) uses B's schedule but keeps the last epoch; B (green) keeps the best dev epoch.

	UUAS (%), punctuation excluded					depth Spearman (NSpr)			
	A, paper schedule		A, const. lr	B, const. lr		A, paper	A, const.	B, const. lr	
ℓ	k=32	k=128	k=128, last ep.	k=32	k=128*	k=32	k=128	k=32	k=128
1	64.5	65.6	50.8	56.1	49.3*	0.733	0.663	0.695	0.666
2	66.4	67.9	50.1	57.8	51.4*	0.757	0.680	0.713	0.647
3	70.6	72.0	52.8	61.7	55.3*	0.777	0.698	0.736	0.685
4	74.8	76.8	58.1	65.5	58.2*	0.799	0.734	0.764	0.712
5	76.6	78.7	60.0	65.5	62.7*	0.824	0.756	0.790	0.709
6	77.6	79.6	59.8	66.7	63.0*	0.836	0.779	0.785	0.718
7	77.7	79.5	60.1	67.8	64.0*	0.845	0.788	0.794	0.753
8	77.4	79.5	62.0	68.9	64.5*	0.855	0.803	0.802	0.757
9	74.5	76.3	59.5	66.3	61.9*	0.846	0.797	0.795	0.749
10	71.3	72.0	57.0	62.9	57.0*	0.826	0.760	0.780	0.715
11	67.8	69.1	55.9	60.7	54.9*	0.812	0.744	0.766	0.712
12	66.4	67.6	58.2	61.9	54.2*	0.814	0.773	0.776	0.752

Table 5: Per-layer test results. *B reports $k = 128$ only with punctuation included. Shaded: ℓ^*=8.

system	ℓ^*	k	UUAS	DSpr	NSpr	Root%
Paper: BERT-base (PTB WSJ, cased)	7 (0-idx.)	full	79.8	0.85	0.87	88.0
A: paper schedule (×0.1 lr decay, best dev)	8	128	79.5	0.822	0.859	87.5
A: constant lr, last epoch (control)	8	128	62.0	0.760	0.803	58.6
B: constant lr, best dev epoch	8	32	68.9	0.784†	0.802‡	72.4

Table 6: Best probes vs. Hewitt and Manning (2019) Table 1. UUAS excludes punctuation in all rows (B with punctuation: 66.4). †B's distance Spearman; ‡B averages per sentence, A per length (the paper's definition).

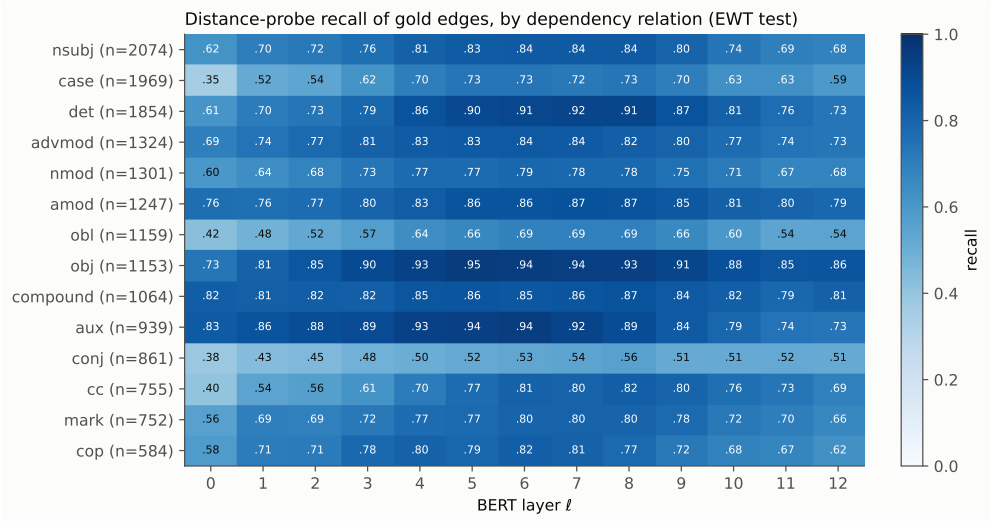


Figure 2: A: probe recall of gold edges per dependency relation and layer (EWT test).

5 Task 3 – BERT-derived static vectors

Method (both)

- Average the layer-8 vectors of every training occurrence of a word type (mean over word pieces, then over occurrences) – the mean/mean pooling that Bommasani et al. (2020) found best – and evaluate with the Task 1 functions.
- A evaluates all 16,654 types; B's main comparison restricts T3 to T1's 8,865 types (all types as a variant), so B's covered sets are smaller.

		WordSim-353 ρ		analogy accuracy (%)	
impl.	system	248 covered	203 (count ≥2)	covered	all 19,544
A	T1 PPMI-SVD	0.104 (248)	0.171	0.51 (5,727)	0.15
B	T1 PPMI-SVD (count ≥2)	–	0.189	0.85 (3,406)	0.15
A	T3 layer 8, all types	0.577 (248)	0.648	67.7 (5,727)	19.8
B	T3 layer 8, all types	0.577 (248)	–	67.7 (5,727)	19.8
B	T3 layer 8, T1 vocabulary (main)	–	0.648	77.1 (3,406)	13.4
A	T3 + all-but-the-top (D=7)	0.682 (248)	0.716	71.0 (5,727)	20.8
B	T3 mean-centred (T1 vocab.)	–	0.706	78.3 (3,406)	13.7
B	T3 centred + PCA-100 (T1 vocab.)	–	0.678	81.4 (3,406)	14.2

Table 7: Semantic benchmarks. The "203" column scores every system on the same 203 pairs (A re-scored; Table 2). B's T1-vocabulary variants also restrict the analogy candidates to 8,865 words, which raises covered accuracy.

	20 sampled sentences		robustness set	
UUAS (%)	A (234 edges)	B (282 edges)	A: 660 sent.	B: 1,926 sent.
random tree	15.2	13.1	17.1	14.3
linear chain	43.2	35.5	43.0	40.5
T1 PPMI-SVD	20.5	16.7	24.8	17.8
T3 layer 8 (A: all types; B: T1 vocab.)	18.4	9.6	17.9	14.5
T3 layer 8, all types	18.4	9.9	17.9	14.7
T3 + anisotropy fix (A: ABTT; B: centred)	19.7	13.5	19.6	16.0
T2 probe, layer 8 (A only)	81.2	–	82.9	–

Table 8: Structural check (UUAS of the cosine-distance MST). The two implementations use different sentences (Table 1), so compare rows within a column.

Meaning: both find a large, significant gain

- **Same 203 pairs:** ρ goes from 0.171 (A) / 0.189 (B) to 0.648 / 0.648.
- **Analogies:** from below 1% to 68–77% on covered questions.
- **Significance (A):** $\Delta\rho$ = +0.474 [0.36, 0.61]; T3 solves 3,848 analogy questions T1 misses, T1 solves 0 that T3 misses (Table 9).
- **Caveat (both):** not a data-matched comparison – BERT brings pre-training on ≈3.3B words; EWT only supplies the contexts.
- **Anisotropy:** removing the shared direction helps in both (A all-but-the-top 0.682; on the 203 pairs, B centring 0.706 vs. A all-but-the-top 0.716), confirming that raw BERT vectors are anisotropic (Ethayarajh, 2019).

Structure: both find no improvement

- **20-sentence samples:** T3 below T1 in both (A 18.4 vs. 20.5; B 9.6 vs. 16.7).
- **Larger sets:** T3 below T1 in both (A 17.9 vs. 24.8; B 14.5 vs. 17.8), close to the random tree.
- **Significance (A):** the 20-sentence difference is not significant (Δ =-2.1 pt, CI [−6.4, 3.2]); the 660-sentence one is (Δ =-6.9 pt, CI [−8.0, −5.8]). B reaches the same conclusion from its larger full-test set.
- **So the two evaluations disagree – in both implementations.**

	WordSim-353	Analogy (McNemar)		MST ΔUUAS (pt) [95% CI]	
A − B	$\Delta\rho$ [95% CI]	only A / only B	p	20 sentences	pool (660)
T3 − T1	+0.474 [0.36, 0.61]	3,848 / 0	$< 10^{-300}$	−2.1 [−6.4, 3.2]	−6.9 [−8.0, −5.8]
T3+ABTT − T1	+0.579 [0.46, 0.71]	4,036 / 0	$< 10^{-300}$	−0.9 [−6.6, 5.6]	−5.1 [−6.2, −4.1]
T3+ABTT − T3	+0.105 [0.05, 0.16]	276 / 88	1×10^{-23}	+1.3 [−3.2, 5.6]	+1.7 [1.0, 2.6]
T2 probe − T3	–	–	–	+62.8 [53.2, 69.6]	+65.0 [63.8, 66.3]

Table 9: A's paired tests on identical items (analogy: questions solved only by A / only by B, exact McNemar p).

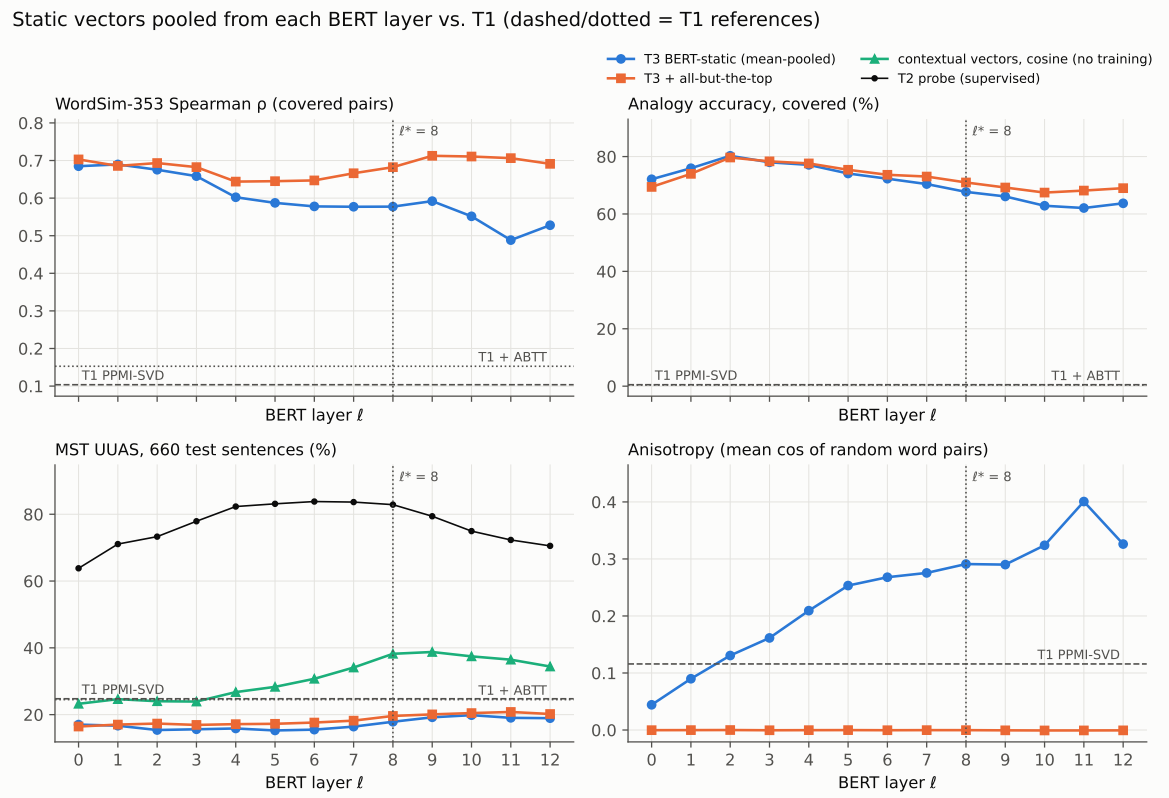


Figure 3: A's layer sweep: static vectors pooled from each layer (raw and all-but-the-top) vs. T1; the MST panel also shows raw contextual vectors and the probe on the same 660 sentences. B's sweep (its Fig. 2) has the same maxima: WordSim at layer 1, analogy at layer 2.

Layer sweep (both): the syntax layer is not the meaning layer (Fig. 3)

- **WordSim peaks at layer 1** in both (A 0.690 on 248 pairs, B 0.723 on 203), matching Bommasani et al. (2020) (layer 1, $\rho = 0.683$ for bert-base-uncased with 10k contexts).
- **Analogies peak at layer 2** in both (A 80.3%, B 86.2%).
- **Best static-MST layer is 10** in both (A 19.8%, B 16.0%) – still close to chance.
- **Anisotropy (A):** raw WordSim falls with depth while anisotropy rises (0.04 → 0.40); after all-but-the-top the curve is almost flat (0.64–0.71), so much of the decline is the "narrow cone" effect.

6 Why the semantic and syntactic answers disagree

B proposes three reasons; A tests each with a measurement (`diagnostics_task3.py`, `extensions_task3.py`).

1. **Pooling removes the context-dependent signal** (B).
   - Measured by A's ladder on the same 660 sentences (Table 10): static T3 17.9% → the same vectors through the probe's matrix B 35.9% → per-token contextual vectors, plain cosine 38.2% → trained probe 82.9%.
2. **The probe learns a projection; the MST check uses raw cosine** (B).
   - Confirmed: even contextual vectors with raw cosine (38.2%) stay below the linear chain (43.0%).
   - The tree is in a learned low-rank subspace, not in the raw geometry.
3. **Edges join semantically dissimilar words; PPMI's window captures local co-occurrence** (B). A measured this as paradigmatic vs. syntagmatic similarity (Sahlgren, 2006) (Table 11):
   - **T3 neighbours are substitutes:** 0.74 share the word's part of speech (T1 0.33; random 0.22).
   - **T1 neighbours are partners:** 0.56 co-occur with the word within ±5 (T3 0.16; random 0.07).
   - **Direct test, edge AUC** (does closeness predict attachment inside a sentence?): T3 0.46 (no information), T1 0.54 (weakly informative), contextual 0.70, probe 0.96.
   - **Refinement:** gold edges are *not* mostly between different parts of speech relative to non-edges (same-UPOS rate 0.13 vs. 0.09), so the effect is about similarity, not word category.

Further observations

- **Hubs (A):** T3's MSTs are star-shaped around frequent function words (degree 2.29 vs. gold 1.05).
- **OOV words (B):** 7.4% of the words in B's sample have no vector and inject arbitrary distances; A's in-vocabulary sample avoids this, and T3 is still below T1.

representation for the MST	tier	20 sentences [95% CI]	pool (660) [95% CI]	edge AUC
random tree (expected)	–	15.2	–	–
linear chain	–	43.2 [40.6, 46.4]	43.0 [42.4, 43.7]	0.75
T1 PPMI-SVD (spec)	req.	20.5 [16.3, 26.2]	24.8 [23.7, 26.0]	0.54
T1 PPMI-SVD (p0)	B	22.6 [17.8, 28.4]	24.5 [23.4, 25.6]	–
T3 BERT-static L8	req.	18.4 [13.6, 26.0]	17.9 [16.8, 18.9]	0.46
T3 BERT-static L8 + ABTT	B	19.7 [14.9, 26.2]	19.6 [18.6, 20.7]	0.44
concat [T1 ; T3] (L2-normalised)	C	19.2 [15.2, 25.1]	25.1 [24.0, 26.2]	–
T3 BERT-static L8 through probe B	C	29.5 [23.7, 38.2]	35.9 [34.3, 37.5]	–
contextual L8, cosine (no training)	B	35.9 [32.1, 39.6]	38.2 [37.3, 39.2]	0.70
T2 distance probe L8 k=128	ref.	81.2 [75.9, 85.9]	82.9 [81.9, 83.9]	0.96

Table 10: A's diagnostic ladder on the same sentences; edge AUC = probability that a gold edge pair is closer than a non-edge pair in the same sentence.

	T1 PPMI-SVD	T3 layer 8	T3 + ABTT	random / gold
NN with same majority UPOS (top-10)	0.326	0.740	0.710	0.223
NN that co-occur within ±5 in train	0.557	0.162	0.156	0.067
hubness: skewness of N_{10}	2.10	1.45	0.57	–
function words among top-50 hubs	0.10	0.12	0.02	–
edge AUC (gold edge closer than non-edge)	0.54	0.46	0.44	0.50
MST: mean degree of function-word tokens	1.84	2.29	2.04	1.05 (gold)
MST: mean max. degree per sentence	3.75	5.46	3.62	4.53 (gold)
MST: edges joining same-UPOS words	0.21	0.25	0.29	0.13 (gold)

Table 11: A's diagnostics: neighbour types, hubness, edge AUC and MST shape.

7 Qualitative analysis

T1 neighbour lists agree across implementations

- Despite different vocabularies and SVD solvers: *monday* → friday, july, tuesday; *good* → nice, food, great, really, pretty; *said* → spokesman, gregg, maviglio, penman; *computer* → disc, evidence, plans, mohammed (all in both).
- Both read them the same way: topical co-occurrence partners from the corpus (reviews, news), not synonyms.
- T3's neighbours are substitutes (Table 12).

word	T1 PPMI-SVD: 5 nearest neighbours	T3 BERT-static (layer 8): 5 nearest neighbours
good	nice, food, pretty, great, really	excellent, great, nice, decent, wonderful
bad	faith, big, too, wrong, something	terrible, horrible, poor, good, shitty
monday	friday, 0832, july, sunday, 6	tuesday, sunday, thursday, wednesday, saturday
dog	got, horse, cat, myself, leave	dogs, cat, puppy, animal, cats
car	home, guy, test, driving, drive	cars, vehicle, vehicles, suv, ferrari
money	save, enough, much, else, attention	cash, funds, dollars, cashout, bucks
said	spokesman, nun, penman, gregg, maviglio	says, stated, saying, told, say
happy	definitely, customer, guys, !!, safe	pleased, glad, unhappy, satisfied, delighted
computer	disc, plans, evidence, read, mohammed	laptop, computers, pc, machine, internet
two	screamed, paddled, expressway, kitchens, revolt	three, four, 2, five, six

Table 12: Nearest neighbours in A (B's T1 lists, given in the text, agree).

Task 2 errors (A; chosen by rank among 10–18-word test sentences, Fig. 4)

- **Median** (UUAS 0.82): "Say after I finished those 2 years and I found a job." – loses the discourse attachment of "Say" and links the two subjects "I" to each other.
- **Worst** (0.30): the verbless itinerary "Raleigh to Chicago, Chicago to Seoul and then Seoul to Manila." – the probe links *repeated identical words* (to–to, Chicago–Chicago).
- **Lesson:** identical word forms get similar BERT vectors, so lexical identity leaks into the syntactic distance.

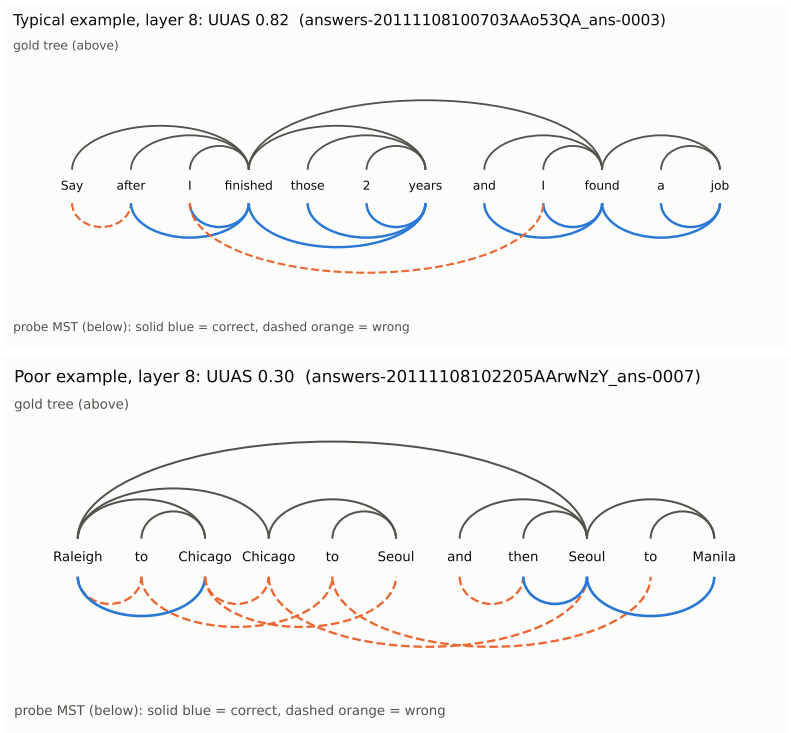


Figure 4: A's probe (layer 8) on the median and worst 10–18-word test sentences. Gold tree above, probe MST below (blue = correct, orange dashed = wrong).

Task 3 analogies (A, Table 13)

- **Successes:** near-perfect on regular morphology (plural 96.6%) and frequent capitals (93.6%).
- **Failures:** currency (0/28; "real" is always an adjective in EWT train).
- **Near misses:** a semantic neighbour beats the morphological target (cooler → nicer, biggest → largest) – the same substitute bias.
- **One representative sentence** of A's 20 (Fig. 5, appendix): T1 and T3 get 25%, raw contextual vectors 33%, the probe 83%; the T3 tree is a star around "a" and "of".

section	question (a : b :: c : **gold**)	T3 answer	T1 answer
capital-common-countries	baghdad : iraq :: beijing : **china**	china ✓	jintao
capital-world	amman : jordan :: baghdad : **iraq**	iraq ✓	threats
city-in-state	houston : texas :: philadelphia : **pennsylvania**	pennsylvania ✓	franchise
family	boy : girl :: brother : **sister**	sister ✓	cousin
gram1-adjective-to-adverb	amazing : amazingly :: complete : **completely**	completely ✓	digest
gram2-opposite	acceptable : unacceptable :: aware : **unaware**	unaware ✓	socially
capital-world	beijing : china :: kingston : **jamaica**	jamaican	northwest
currency	brazil : real :: canada : **dollar**	actual	church
gram2-opposite	acceptable : unacceptable :: certain : **uncertain**	specific	socially
gram3-comparative	bad : worse :: cool : **cooler**	nicer	dried
gram4-superlative	bad : worst :: big : **biggest**	largest	chinese
gram6-nationality-adjective	australia : australian :: england : **english**	british	soper

Table 13: A: analogy questions T3 solves and T1 does not (top) and T3 failures (bottom).

8 Discussion

What the comparison adds

- Two independent implementations make the main claims much stronger: the best syntax layer, the T3 semantic gain, the T3 structural failure, the disagreement between the evaluations and the best static layers are all reproduced.
- The Task 3 numbers match exactly once settings are aligned.
- **Three choices change numbers by more than the effects being studied** – reports should state them explicitly:
  1. the probe's learning-rate schedule (11–18 pt UUAS; it also flips the best rank);
  2. the evaluation subset (248 vs. 203 covered pairs move T1's ρ from 0.10 to 0.17–0.19; the MST sample moves the baselines by up to 8 pt);
  3. the OOV imputation rule for "all 353" (0.048 vs. 0.060 for A's own vectors).

What worked / what did not

- **Worked:** the probe reproduces the paper when its schedule is reproduced; pooled BERT vectors are strong lexical vectors; one post-processing step (centring or all-but-the-top) helps.
- **Did not work:** PPMI-SVD on 200k words cannot do analogies; pooling from the *syntax* layer does not produce *syntactic* static vectors – the semantic benchmarks measure similarity between word types, while trees are relations between tokens in context, which pooling removes.

Limitations

- B's numbers are transcribed from B's report (no raw outputs); B's DSpr/NSpr definitions differ slightly from A's.
- Each probe was trained once; the two MST samples differ.
- BERT vs. PPMI is not a data-matched comparison; one treebank.

9 Reproducibility

- **A:** `SEED=42` (Python, NumPy, PyTorch) for the sample, baselines, bootstraps and SVD; probe seed $42 + 1000\ell + k$ (+500 depth); pinned EWT tag, gensim commit and model revision `86b5e09`; `python code/run_all.py` or the Colab notebook runs everything; every number is generated from the outputs; 2×L40S, 78 probes in 49+54 min.
- **B:** sentence-sample seed 42, SVD start vector 0, probe seed $1000k$ (+7 depth) + first layer of the chunk; EWT `.conllu` files; Task 1 in about 13 s on CPU.
- **Code:** https://github.com/trishitasaha/NLP_Assignment2 – B at the root (`Codes/`, `results/`), A in `23CS10081_Yellapu_Dheeraj_Babu/`, this report in `Combined/`.
- **Deviations from the specification:**
  - both: best-dev checkpoint selection; ℓ^* chosen on dev, not test; WS-353 and analogies reported on covered and on all items; the 20-sentence check complemented by a larger set;
  - A only: the paper's lr decay (justified by the control in §4); layer 0 probed; punctuation excluded from the structural check.

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A Additional material (implementation A)

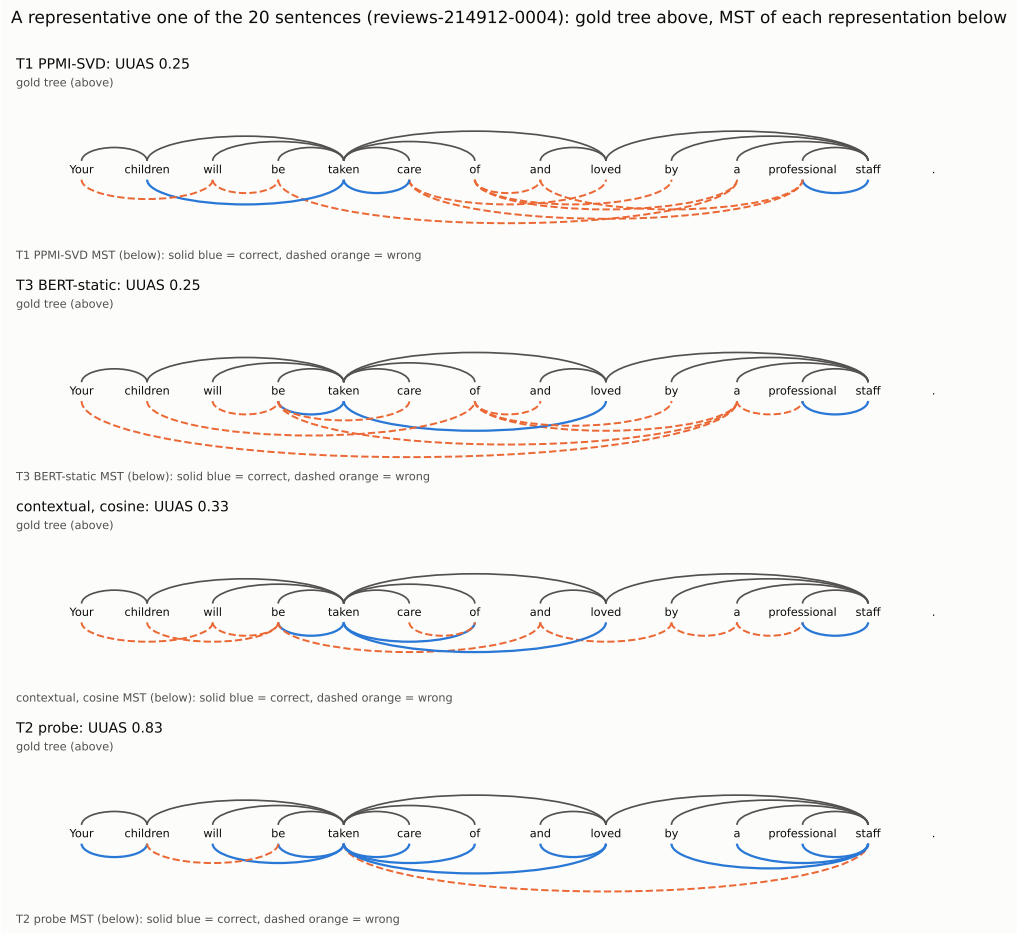


Figure 5: "Your children will be taken care of and loved by a professional staff." – gold tree (above) and the MST of each representation (below), implementation A.

	WS-353 ρ (cov.)		Analogy (cov., %)		MST UUAS on the pool (%)				
ℓ	raw	ABTT	raw	ABTT	static raw	static ABTT	contextual cos.	T2 probe	anisotropy
0	0.685	0.703	72.1	69.4	17.0	16.4	23.2	63.8	0.04
1	0.690	0.686	75.9	74.0	16.7	17.0	24.6	71.1	0.09
2	0.675	0.693	80.3	79.7	15.4	17.3	24.0	73.3	0.13
3	0.658	0.682	78.0	78.3	15.6	16.9	23.9	77.9	0.16
4	0.602	0.644	77.1	77.6	15.9	17.2	26.7	82.3	0.21
5	0.587	0.645	74.1	75.4	15.3	17.3	28.3	83.1	0.25
6	0.578	0.647	72.3	73.7	15.5	17.6	30.7	83.8	0.27
7	0.577	0.666	70.4	73.0	16.4	18.2	34.1	83.6	0.28
8	0.577	0.682	67.7	71.0	17.9	19.6	38.2	82.9	0.29
9	0.592	0.713	66.1	69.2	19.2	20.1	38.8	79.4	0.29
10	0.552	0.711	62.9	67.5	19.8	20.5	37.4	75.0	0.32
11	0.488	0.706	62.1	68.2	19.0	20.8	36.5	72.3	0.40
12	0.528	0.691	63.7	69.0	18.9	20.2	34.4	70.5	0.33
T1	0.104	0.153	0.5	0.4	24.8	24.5	–	–	0.12

Table 14: A's layer sweep in numbers (covered items; pool = 660 sentences).

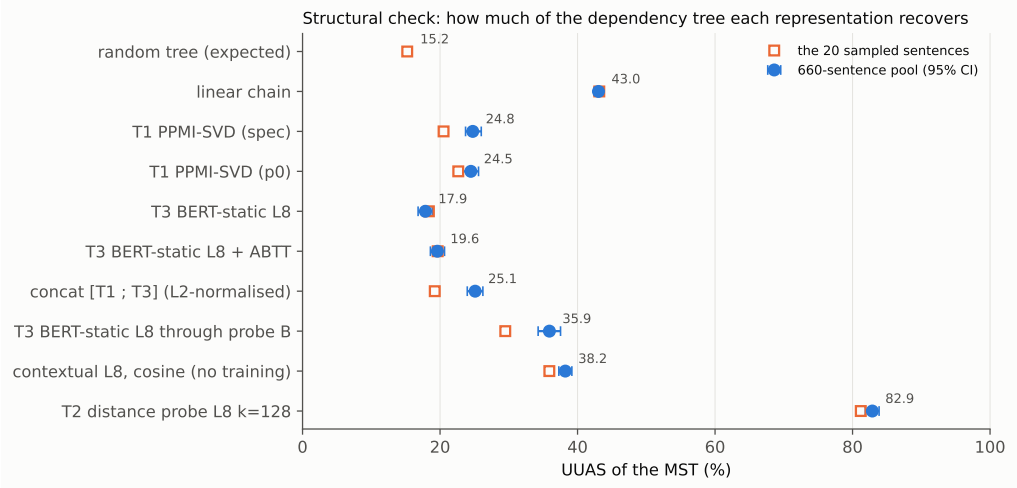


Figure 6: A's diagnostic ladder: MST UUAS on the 660-sentence pool (95% CI) and on the 20 sentences.